Understanding Support Vector Machines and Kernel Functions

1. Introduction

Support Vector Machines (SVM) are among the most powerful supervised machine learning algorithms used for classification and regression tasks. They are particularly effective for analysing high■dimensional numerical data and have been widely used in fields such as healthcare, finance, bioinformatics, and image recognition. The main idea behind SVM is to find an optimal decision boundary that separates classes while maximising the margin between them. A larger margin typically improves the model’s ability to generalise to new unseen data. However, many real■world datasets are not linearly separable. To overcome this limitation, SVM uses kernel functions which transform the input data into higher dimensional spaces. This allows the algorithm to learn nonlinear decision boundaries. This tutorial explains how different kernel functions affect the performance and behaviour of SVM models.

2. SVM Fundamentals

SVM attempts to identify a hyperplane that separates data points belonging to different classes. In two dimensions this hyperplane is simply a line, while in higher dimensions it becomes a plane or hyperplane. The optimal hyperplane is defined as the one that maximises the margin between the decision boundary and the closest points from each class. These closest points are known as support vectors. Because only support vectors influence the model, SVMs can be robust to noise and perform well even when the dataset contains many irrelevant features.

3. The Kernel Trick

Many datasets cannot be separated using a simple straight line. Instead of explicitly transforming the dataset into a higher dimensional space, SVM uses a technique known as the kernel trick. A kernel function computes similarity between data points in a higher dimensional feature space without explicitly performing the transformation. This significantly reduces computational cost.

4. Kernel Types

Common kernel functions include: Linear Kernel – simple and computationally efficient. Polynomial Kernel – captures interactions between features. Radial Basis Function (RBF) Kernel – very flexible and capable of modelling complex nonlinear relationships. Sigmoid Kernel – inspired by neural network activation functions.

5. Dataset

This tutorial uses the Breast Cancer Wisconsin dataset available in the Scikit■learn library. The dataset contains 569 observations and 30 numerical features describing characteristics of cell nuclei extracted from breast cancer biopsies. The classification task is to determine whether a tumour is benign or malignant.

6. Methodology

The following workflow is used: 1. Load the dataset 2. Split into training and testing data 3. Apply feature scaling 4. Train SVM models using multiple kernels 5. Compare performance 6. Optimise hyperparameters using GridSearchCV

Kernel Comparison Table

Kernel	Typical Accuracy	Strength	Weakness
Linear	High	Fast and simple	Limited nonlinear modelling
Polynomial	Moderate■High	Captures feature interactions	Risk of overfitting
RBF	Very High	Flexible nonlinear boundaries	Requires tuning
Sigmoid	Moderate	Neural network behaviour	Less stable

7. Kernel Performance

Experiments show that the RBF kernel often produces the highest accuracy because it can capture complex nonlinear patterns. Linear kernels perform well when the dataset is nearly linearly separable.

8. Hyperparameter Optimisation

Hyperparameters such as C and gamma significantly influence SVM performance. GridSearchCV evaluates multiple parameter combinations using cross■validation to determine the optimal configuration.

9. Ethical Considerations

Machine learning systems used in healthcare must be carefully evaluated. Bias in datasets may cause models to perform poorly for underrepresented groups. AI should support medical professionals rather than replace them.

10. Conclusion

Support Vector Machines remain a powerful classical machine learning algorithm. Understanding kernel functions allows practitioners to adapt SVM models to a wide range of classification problems.

References

- Cortes & Vapnik (1995) Support■Vector Networks
- Bishop (2006) Pattern Recognition and Machine Learning
- Hastie et al. (2009) Elements of Statistical Learning